

COM6115 Sentiment Analysis Coursework Report

Introduction

This report studies sentiment classification with three techniques: a Naïve Bayes classifier that was trained on the Rotten Tomatoes (RT) polarity dataset, a dictionary-based rule system and an extended rule-based model that handles negation and intensity. Evaluation on in-domain film review and cross-domain generalisation to Nokia product review. Comparison between Statistical and Lexicon-based approach is discussed, most informative features for Naive Bayes are shown and detailed error analysis highlighting model-specific failure patterns is presented.

Step 2 - Naïve Bayes on Rotten Tomatoes

Sentiment dictionaries are preprocessed by reading each file line by line and then removing the empty lines as well as filter out all comments starting with “;” . The rest were taken as valid sentiment words. I also did not change the case, nor removed + - signs since these don't occur in the dataset after tokenising so they shouldn't affect classification. This way the lexicon gets loaded cleanly, and we retain all valid sentiment entries.

Naive Bayes got around 0.78 accuracy on the RT test set, which is close to what such models usually score. Precision and recall for both classes are balanced ($\approx 0.77 - 0.78$) meaning that our classifier is not favoring either of the classes as the data is balanced. F1-score of the positive class (≈ 0.78) means that although Naïve Bayes is a simple model and makes independent assumptions, it does capture the main vocabulary that contains sentiments in RT reviews. The training/test gap (≈ 0.89 vs 0.77) shows natural overfitting to the training distribution.

Step 3 - Cross-Domain Evaluation

The classifier achieves good performance on the training set (≈ 0.89 accuracy) but falls to around 0.77 on the RT test set, indicating decent generalisation within the same domain. However, the performance degrades significantly for the Nokia product review domain and is around 0.59. This is a very clear domain shift. The model has learned the probability distribution of film-review words (plot, acting,

boring, thrilling). Nokia reviews use a completely different lexicon of words associated with device quality (e.g. battery, signal, camera, charging). Thus, most of the words were unseen during training and their likelihood estimate is inaccurate.

Also, there is imbalance cross-domain. On the Nokia dataset, Naïve Bayes has high precision but low recall for positive review and very low precision for negative review. This suggests a very strong tendency to label a lot of negative Nokia reviews as positive. words like problem, screen, signal or battery may be sentiment-bearing in a product-quality context, but are neutral or rare in film reviews, so the model gets them wrong. In general, the large gap between in-domain and out-of-domain performance shows that Naïve Bayes is dependent on word distributions specific to a particular domain and fragile to different vocabularies and contexts than the training data.

Table 1 — Naïve Bayes Results (Films + Nokia)

Dataset	Accuracy	Precision (Pos)	Recall (Pos)	Precision (Neg)	Recall (Neg)	F1 (Pos)
Films (Train)	0. 8899	0. 8985	0. 8792	0. 8818	0. 9007	0. 8887
Films (Test)	0. 7696	0. 7771	0. 7566	0. 7624	0. 7826	0. 7667
Nokia	0. 5902	0. 7984	0. 5538	0. 3942	0. 6750	0. 6540

Step 4 – Most Informative Words

Among the most useful predictors that Naïve Bayes learns, we find both standard sentiment terms as well as more domain specific evaluative expressions. Among the negative words, items such as boring, unfunny, pathetic, lousy, and dull correspond to clearly negative evaluative language and appear in the sentiment dictionary. But it also finds domain-specific clues like seagal, chan, pokémon, wannabe and product, these are not general clues to sentiment, but occur often in negative reviews of films. This shows that Naïve Bayes models emotional polarity as well as contextual indicators of low-quality films.

Positive predictors include beauty, wonderful, vibrant, touching, warm, and delightful both in the dictionary and among the top learned features. Others—nuanced, evocative, captivating, intimate, realistic, engrossing—are domain-specific quality descriptors which are not found in the lexicon, yet indicate a positive response in film criticism. About half the top negative and two thirds the top positive are in the lexicon. That shows that though Naive Bayes does well at most of the

usual sentiment cues, it is really dependent on domain specific evaluative vocab that's not in general purpose lexicons.

Step 5 - Dictionary Methods and Improved Rules

5.1 Baseline dictionary vs Naïve Bayes

The dictionary-based classifier is substantially worse than Naïve Bayes within the RT film domain. its training and test accuracies (≈ 0.65 and ≈ 0.68) are much lower than those of Naïve Bayes (≈ 0.89 and ≈ 0.77). which is expected, as the dictionary method uses a static lexicon and does not learn context- or domain-specific sentiment cues. Film reviews employ lots of stylistic and evaluative words and expressions like cliched, uninspired, nuanced, riveting, which aren't in the lexicon. Naïve Bayes, however, learns them directly from the data, which gives it better in-domain performance.

Table 2 — Dictionary Baseline Results

Dataset	Accuracy	Precision (Pos)	Recall (Pos)	Precision (Neg)	Recall (Neg)	F1 (Pos)
Films (Train)	0.6492	0.6792	0.5670	0.6278	0.7317	0.6180
Films (Test)	0.6808	0.7099	0.6015	0.6598	0.7587	0.6512
Nokia	0.7970	0.8837	0.8172	0.6383	0.7500	0.8492

5.2 Improved rule-based classifier

I added to the dictionary three types of linguistic rules:

- (1) negation (e.g., not good $\rightarrow$ polarity reversed),
- (2) intensifiers (e.g., very good $\rightarrow$ increased polarity magnitude),
- (3) diminishers (e.g., slightly bad $\rightarrow$ weakened polarity). I used the token preceding each sentiment word and adjusted its score.

On the RT test set, the improved system gives only a small improvement ($\approx 0.655 \rightarrow \approx 0.661$). This limited gain shows that film reviews are very heavy on domain specific evaluative vocabulary; the rules can fix some local syntactic patterns but this

is not enough to make up for the missing lexical coverage.

On the other hand, on the Nokia dataset it is more apparent ($\approx 0.797 \rightarrow \approx 0.808$). The product reviews also use simpler and more literal sentiment constructions (not good, very bad, barely acceptable). Therefore, the rule extensions are more aligned with the actual sentiment expressions in this domain.

Table 3 — Improved Dictionary (Negation + Intensifier + Diminisher)

Dataset	Accuracy	Precision (Pos)	Recall (Pos)	Precision (Neg)	Recall (Neg)	F1 (Pos)
Films (Train)	0.6556	0.6864	0.5768	0.6331	0.7349	0.6268
Films (Test)	0.6611	0.6832	0.5610	0.6465	0.7553	0.6161
Nokia	0.8083	0.8994	0.8172	0.6495	0.7875	0.8563

5.3 Summary

Naive Bayes is good within the training domain, but not for transfer.

Dictionary-based is worse in-domain, but much more robust across-domains. The improved rule-based variant gave some small gains on films and clear benefits on Nokia reviews, which had simpler ways to express sentiment.

Step 6 – Error Analysis

The misclassified sentences from Naive Bayes show some systematic tendencies. The classifier frequently fails on mixed-polarity and contrastive constructions: effective but too-tepid biopic thought-provoking. but it pays a price. It because it assumes that words are independent, it can't model discourse markers like “but”, which are often used to indicate the reviewer has started talking about their actual opinion. and it gets really confused on stuff that is not direct and is not actually what they are talking about. For instance, when someone is saying good things but uses the road to hell, it gets lost. Some examples show that Naïve Bayes misinterpretation of more evaluative language such as a beautiful and haunting examination. Where the single negative words carry a more weight than the positive overall statement. Finally, the model misclassifies sentences with evaluative terms specific to a certain domain, which may be rare in the training data and thus poorly estimated.

Dictionary based classifier has different type of errors. Most are from its failure to deal with negation like “not find what they’re looking for” or “never funny,” which have positive surface words but mean negative things. The method also fails to treat intensifiers and diminishers appropriately, treating “very bad” and “barely interesting” as having the same polarity strength. Lots of errors come from not having the right words that are special to a field, like pretentious, indulgent, or inconsequential, so it’s hard to give good scores for how someone feels about a movie. And the dictionary method also wrongly classifies certain types of compound expressions (like “a not-great movie”) as positive because it has some positive tokens in it.

Overall naive bayes failed due to context model, but not on discourse model. Dictionary base failed due to not all words are covered on the lexicon and not have structuring on syntactic and semantic. The two thus make different kinds of errors.

Conclusion

In this study, we can see that Naïve Bayes is effective for the in-domain film reviews, but it does not generalize well to other domains. The dictionary based classifier has poor performances w.r.t to the films since there are fewer occurrences of words but is by far more robust over all domains. Applying some simple linguistic rules to the lexicon-based method makes it better, especially when talking about things that have the same way of expressing good or bad feelings in those things. In the comparison part, it presents a trade-off between data-driven and Lexicon based. Also shows the nature of the domain is important to the sentimental analysis.

Appendix A — Most Useful Words Learned by Naïve Bayes

NEGATIVE: ['unfunny', 'routine', 'waste', 'generic', 'badly', 'mediocre', 'poorly', 'mindless', 'boring', 'pointless', 'disguise', 'stale', "wasn't", 'offensive', 'shoot', 'bore', 'meandering', 'apparently', 'tiresome', 'stupid', 'chan', 'annoying', 'plodding', 'pinocchio',

'unless', 'seagal', 'harvard', 'flat', 'ill', 'stealing', 'amateurish',
'horrible', 'kung', 'wasted', 'sadly', 'pokemon', 'product', 'ballistic',
'uninspired', 'conceived', 'lifeless', 'dull', 'supposed', 'inept',
'collection', 'lame', 'animal', 'pathetic', 'trite', 'cliched',
'unintentional', 'vampire', 'incoherent', 'lousy', 'choppy', 'leaden',
'soggy', 'junk', 'busy', 'halfway', 'muddled', 'store', 'cliche', 'gag',
'pow', 'generate', 'paper', 'wannabe', 'crap', 'stunt', 'tuxedo',
'unimaginative', 'sara', 'unnecessary', 'numbers', 'trapped', 'britney',
'ship', 'unintentionally', 'bother', 'serving', 'cable', 'banal',
'settles', 'arts', 'flash', 'exploitative', 'witless', 'wilson',
'missed', 'writers', 'putting', 'pile', 'schneider', 'comparison',
'thrown', 'pie', 'named', 'dogs', 'lackluster']

POSITIVE: ['gently', 'straightforward', 'finely', 'para', 'color',
'resist', 'beauty', 'evocative', 'masterful', 'bringing', 'buoyant',
'discovery', 'lovers', 'gut', 'format', 'understands', 'poem', 'flaws',
'affecting', 'speaks', 'deadpan', 'pianist', 'transcends', 'integrity',
'exquisitely', 'washington', 'nuanced', 'record', 'delightfully',
'superbly', 'answers', 'tasty', 'spite', 'hopeful', 'devastating',
'reminds', 'portrayal', 'intoxicating', 'joyous', 'touching', 'unique',
'lane', 'bourne', 'wrenching', 'russian', 'potent', 'aware',
'captivating', 'unflinching', 'martha', 'frailty', 'sly', 'gradually',
'explores', 'richly', 'timely', 'smarter', 'sadness', 'warm', 'pulls',
'playful', 'startling', 'powerful', 'aspects', 'unexpected', 'spare',
'heartbreaking', 'grown', 'vibrant', 'wry', 'vividly', 'literary',
'subversive', 'jealousy', 'iranian', 'detailed', 'resonant', 'sides',
'polished', 'wonderfully', 'provides', 'tender', 'lively', 'tour',
'respect', 'heartwarming', 'realistic', 'gem', 'mesmerizing',
'wonderful', 'chilling', 'refreshingly', 'ages', 'riveting',
'inventive', 'refreshing', 'absorbing', 'intimate', 'captures',
'engrossing']

Appendix B — Error Samples (Naïve Bayes)

ERROR (pos classed as neg 0.10):effective but too-tepid biopic

ERROR (pos classed as neg 0.02):perhaps no picture ever made has more
literally showed that the road to hell is paved with good intentions .

ERROR (pos classed as neg 0.19):scores a few points for doing what it does
with a dedicated and good-hearted professionalism .

ERROR (pos classed as neg 0.31):some actors have so much charisma that
you'd be happy to listen to them reading the phone book . hugh grant and
sandra bullock are two such likeable actors .

ERROR (pos classed as neg 0.48):a few artsy flourishes aside , narc is
as gritty as a movie gets these days .

ERROR (pos classed as neg 0.30):if it's possible for a sequel to outshine

the original , then sl2 does just that .

ERROR (pos classed as neg 0.01):maybe it's just because this past year has seen the release of some of the worst film comedies in decades . . . but honestly , analyze that really isn't all that bad .

ERROR (pos classed as neg 0.26):britney has been delivered to the big screen safe and sound , the way we like our 20-year-old superstar girls to travel on the fame freeway .

ERROR (pos classed as neg 0.06):my goodness , queen latifah has a lot to offer and she seemed to have no problem flaunting her natural gifts . she must have a very strong back .

ERROR (pos classed as neg 0.33):you might not buy the ideas . but you'll definitely want the t-shirt .

ERROR (pos classed as neg 0.41):an impressive debut for first-time writer-director mark romanek , especially considering his background is in music video .

ERROR (pos classed as neg 0.34):this is such a dazzlingly self-assured directorial debut that it's hard to know what to praise first .

ERROR (pos classed as neg 0.44):the tug of war that ensues is as much a snapshot of modern china in microcosm as it is a crash course in movie mythology .

ERROR (pos classed as neg 0.04):asia authors herself as anna battista , an italian superstar and aspiring directress who just happens to be her own worst enemy .

ERROR (pos classed as neg 0.44):hands down the year's most thought-provoking film . but it pays a price for its intricate intellectual gamesmanship .

ERROR (pos classed as neg 0.28):a model of what films like this should be like .

ERROR (neg classed as pos 0.59):if only merchant paid more attention the story .

ERROR (neg classed as pos 0.86):sex with strangers will shock many with its unblinking frankness . but what is missing from it all is a moral . what is the filmmakers' point ? why did they deem it necessary to document all this emotional misery ?

ERROR (neg classed as pos 0.50):goldmember has none of the visual wit of the previous pictures , and it looks as though jay roach directed the film from the back of a taxicab .

ERROR (neg classed as pos 0.76):despite some charm and heart , this quirky soccer import is forgettable

ERROR (neg classed as pos 0.69):so brisk is wang's pacing that none of the excellent cast are given air to breathe .

ERROR (neg classed as pos 0.56):'cq may one day be fondly remembered as roman coppola's brief pretentious period before going on to other films that actually tell a story worth caring about

ERROR (neg classed as pos 0.74):has a customarily jovial air but a deficit of flim-flam inventiveness .

ERROR (neg classed as pos 0.89):the very definition of what critics have come to term an " ambitious failure . "

ERROR (neg classed as pos 0.72):the soul-searching deliberateness of the film , although leavened nicely with dry absurdist wit , eventually becomes too heavy for the plot .

ERROR (neg classed as pos 0.52):it's a very sincere work , but it would be better as a diary or documenta

Appendix C — Error Samples (Dictionary-Based)

ERROR (pos→neg): jacquot's toska is a treat .

ERROR (pos→neg): murderous maids may well be the most comprehensive of these films and also strike closest to the truth .

ERROR (pos→neg): sure , it's more of the same , but as the film proves , that's not always a bad thing .

ERROR (pos→neg): we need [moore's] noisy , cocky energy , his passion and class consciousness ; we need his shticks , we need his stones .

ERROR (pos→neg): the path ice age follows most closely , though , is the one established by warner bros . giant chuck jones , who died a matter of weeks before the movie's release .

ERROR (pos→neg): a beautiful and haunting examination of the stories we tell ourselves to make sense of the mundane horrors of the world .

ERROR (pos→neg): just another fish-out-of-water story that barely stays afloat .

ERROR (pos→neg): has a shambling charm . . . a cheerfully inconsequential diversion .

ERROR (pos→neg): an incredibly clever and superbly paced caper filled with scams within scams within scams .

ERROR (pos→neg): [fessenden] is much more into ambiguity and creating mood than he is for on screen thrills

ERROR (pos→neg): the comic performances are all spot on , especially lee ross's turn as ken .

ERROR (pos→neg): offers an unusual opportunity to observe the inequities in the death penalty , not just the inherent immorality but also the haphazard administration of it and public misperception of how the whole thing works .

ERROR (pos→neg): a recent favourite at sundance , this white-trash satire will inspire the affection of even those unlucky people who never owned a cassette of def leppard's pyromania .

ERROR (pos→neg): the 3d images only enhance the film's otherworldly quality , giving it a strange combo of you-are-there closeness with the disorienting unreality of the seemingly broken-down fourth wall of the movie screen .

ERROR (pos→neg): . . . an inviting piece of film .

ERROR (pos→neg): you might not want to hang out with samantha , but you'll probably see a bit of yourself in her unfinished story .

ERROR (pos→neg): enjoy it for what it is ; you can hate yourself later .

ERROR (pos→neg): más sarcástica , divertida y demencial que su predecesora , es un buen ejemplo de lo que es el cine de entretenimiento puro y sin complejos .

ERROR (pos→neg): a real story about real people living their lives concerned about the future of an elderly , mentally handicapped family member .

ERROR (pos→neg): often moving and explores the discomfort inherent in the contacts between the american 'hosts' and their 'guests . ' ERROR (pos→neg): if mr . zhang's subject matter is , to some degree at least , quintessentially american , his approach to storytelling might be called iranian .

ERROR (pos→neg): a compelling portrait of moral emptiness ERROR (pos→neg): novak manages to capture a cruelly hilarious vein of black comedy in the situation with his cast of non-actors and a gritty , no-budget approach .

ERROR (pos→neg): [hawn's character]is so bluntly written , without a trace of sentimentality , and so blisteringly defined , that every other character seems overlooked and underwritten .

ERROR (pos→neg): a film that is a portrait of grace in an imperfect world .

ERROR (pos→neg): the film is painfully authentic , and the performances of the young players are utterly convincing .

ERROR (pos→neg): its gross-out gags and colorful set pieces . . . are of course stultifyingly contrived and too stylized by half . still , it gets the job done -- a sleepy afternoon rental .

ERROR (pos→neg): it moves quickly , adroitly , and without fuss ; it doesn't give you time to reflect on

ERROR (neg→pos): even horror fans will most likely not find what they're seeking with trouble every day ; the movie lacks both thrills and humor .

ERROR (neg→pos): tadpole is emblematic of the witless ageism afflicting films : young is cool , and too young is too cool .

ERROR (neg→pos): culkin exudes none of the charm or charisma that might keep a more general audience even vaguely interested in his bratty character .

ERROR (neg→pos): if you liked the 1982 film then , you'll still like it now .

ERROR (neg→pos): instead of accurately accounting a terrible true story , the film's more determined to become the next texas chainsaw massacre . but what about the countless other people who'd merely like to watch a solid tale about a universally interesting soul ?

ERROR (neg→pos): roger michell , who did an appealing job directing persuasion and notting hill in england , gets too artsy in his american debut .

ERROR (neg→pos): an afterschool special without the courage of its convictions .

ERROR (neg→pos): where last time jokes flowed out of cho's life story , which provided an engrossing dramatic through line , here the comedian hides behind obviously constructed routines .

ERROR (neg→pos): it's never a good sign when a film's star spends the entirety of the film in a coma . it's a worse sign when you begin to envy her condition .

ERROR (neg→pos): a great idea becomes a not-great movie . ERROR (neg→pos): sunshine state lacks the kind of dynamic that limbo offers , and in some ways is a rather indulgent piece .

ERROR (neg→pos): even legends like alfred hitchcock and john huston occasionally directed trifles . . . so it's no surprise to see a world-class filmmaker like zhang yimou behind the camera for a yarn that's ultimately rather inconsequential .

ERROR (neg→pos): beers , who , when she's given the right lines , can charm the paint off the wall . . . [but] the script goes wrong at several key junctures .

ERROR (neg→pos): the best you can say about it is it's so uninspired , it barely gives one pause when considering some of the other dreck out there right now .

ERROR (neg→pos): a fairly by-the-books blend of action and romance with sprinklings of intentional and unintentional comedy .

ERROR (neg→pos): blessed with immense physical prowess he may well be , but ahola is simply not an a